

Practice 1 — Gaussians, Conditional Expectation, and KL

Mathematical Foundations of Diffusion Models
Monsoon 2026 · IIT Jammu

Worked derivations and exercises. Not graded, not submitted

What this sheet is for

A generative model has to recover something it cannot see — a clean signal — from something it can: that signal with noise added. Written out, the machinery is a prior, a likelihood, a posterior, and a conditional mean, and it is the same machinery whether the hidden quantity takes two values or lives in $\mathbb{R}^{784}$.

This sheet installs the three computations that machinery runs on when the noise is Gaussian. Nothing here is deep; all of it is used constantly from Module 3 onward, and each piece is derived in full rather than quoted.

Along the way we install the two other computations that Module 3 runs on: the Gaussian composition rule that produces $q(x_t | x_0)$ in closed form, and the KL-between-Gaussians identity that turns the DDPM objective into a squared error.

The concept

Take a discrete hidden cause and a discrete observation, recovered by Bayes. Real data is continuous and the noise is Gaussian, so three things must be installed before the same pipeline can run:

- **§6** — what a Gaussian does under scaling and addition, so that “add noise t times” collapses to a single formula;
- **§7** — that a conditional mean is what squared-error training produces, and what it looks like when the table becomes a Gaussian;
- **§8** — how to measure the gap between two Gaussians, which turns out to be a squared error.

Together they are essentially all the mathematics in a DDPM.

6. Gaussian algebra, ending in the composition rule

6.1 Affine maps

Claim. If $X \sim \mathcal{N}(\mu, \sigma^2)$ and $a \neq 0$, then $aX + b \sim \mathcal{N}(a\mu + b, a^2\sigma^2)$.

Derivation by change of variables (take $a > 0$; $a < 0$ differs only by $|a|$). Start from the CDF, because that is where a monotone map is easiest:

$$F_Y(y) = \mathbb{P}(aX + b \leq y) = \mathbb{P}\left(X \leq \frac{y-b}{a}\right) = F_X\left(\frac{y-b}{a}\right).$$

Differentiate, picking up $1/a$ from the chain rule:

$$f_Y(y) = \frac{1}{a} f_X\left(\frac{y-b}{a}\right) = \frac{1}{a\sqrt{2\pi}\sigma} \exp\left(-\frac{\left(\frac{y-b}{a} - \mu\right)^2}{2\sigma^2}\right).$$

Now clear the inner fraction — the only step with any content:

$$\frac{\left(\frac{y-b}{a} - \mu\right)^2}{\sigma^2} = \frac{(y-b-a\mu)^2}{a^2\sigma^2},$$

so f_Y is a Gaussian density with mean $a\mu + b$ and variance $a^2\sigma^2$. ■

The consequence used constantly: $X = \mu + \sigma Z$ with $Z \sim \mathcal{N}(0, 1)$. Every Gaussian is standard noise, scaled and shifted — the *reparametrisation trick*, and the reason a sampler only ever needs $\mathcal{N}(0, 1)$.

6.2 Sums of independent Gaussians

Claim. $X \sim \mathcal{N}(\mu_1, \sigma_1^2)$ and $Y \sim \mathcal{N}(\mu_2, \sigma_2^2)$ independent $\Rightarrow X + Y \sim \mathcal{N}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.

Derivation by MGF, the short route, set up in the W03 deck. For independent variables the MGF of a sum is the product of the MGFs:

$$M_{X+Y}(t) = M_X(t) M_Y(t) = e^{\mu_1 t + \frac{1}{2}\sigma_1^2 t^2} e^{\mu_2 t + \frac{1}{2}\sigma_2^2 t^2} = e^{(\mu_1 + \mu_2)t + \frac{1}{2}(\sigma_1^2 + \sigma_2^2)t^2}.$$

That is the Gaussian MGF with mean $\mu_1 + \mu_2$ and variance $\sigma_1^2 + \sigma_2^2$; by the uniqueness theorem the distribution is that Gaussian. ■

6.3 The composition — the payoff

The single most load-bearing computation in the course, and it is two lines.

Setup. One noising step, with x_0 held fixed and $\varepsilon \sim \mathcal{N}(0, 1)$:

$$x_1 = \sqrt{\alpha_1} x_0 + \sqrt{1 - \alpha_1} \varepsilon_1, \quad x_2 = \sqrt{\alpha_2} x_1 + \sqrt{1 - \alpha_2} \varepsilon_2,$$

with $\varepsilon_1, \varepsilon_2$ independent. **What is x_2 in terms of x_0 ?**

Substitute the first into the second:

$$\begin{aligned} x_2 &= \sqrt{\alpha_2} \left(\sqrt{\alpha_1} x_0 + \sqrt{1 - \alpha_1} \varepsilon_1 \right) + \sqrt{1 - \alpha_2} \varepsilon_2 \\ x_2 &= \sqrt{\alpha_1 \alpha_2} x_0 + \underbrace{\sqrt{\alpha_2(1 - \alpha_1)} \varepsilon_1 + \sqrt{1 - \alpha_2} \varepsilon_2}_{\text{two independent Gaussians, mean 0}} \end{aligned}$$

Add the variances, by §6.1 for the scalings and §6.2 for the sum:

$$\alpha_2(1 - \alpha_1) + (1 - \alpha_2) = \alpha_2 - \alpha_1 \alpha_2 + 1 - \alpha_2 = 1 - \alpha_1 \alpha_2.$$

The α_2 terms cancel exactly, and therefore

$$\boxed{x_2 = \sqrt{\alpha_1 \alpha_2} x_0 + \sqrt{1 - \alpha_1 \alpha_2} \bar{\varepsilon}, \quad \bar{\varepsilon} \sim \mathcal{N}(0, 1).}$$

Two noising steps have the form of one noising step, with $\alpha_1 \alpha_2$ in place of α . By induction, writing $\bar{\alpha}_t = \prod_{s \leq t} \alpha_s$:

$$\boxed{x_t = \sqrt{\bar{\alpha}_t} x_0 + \sqrt{1 - \bar{\alpha}_t} \varepsilon, \quad \text{i.e.} \quad q(x_t | x_0) = \mathcal{N}(\sqrt{\bar{\alpha}_t} x_0, (1 - \bar{\alpha}_t)I).}$$

The concept

Three things to draw out of that formula.

The parametrisation was designed for this. The pair $(\sqrt{\alpha}, \sqrt{1-\alpha})$ is chosen precisely so the noise variances telescope to $1 - \bar{\alpha}_t$. Any other choice still gives a Gaussian; this one gives a *tidy* Gaussian.

Total variance is preserved. If x_0 has unit variance then x_t has variance $\bar{\alpha}_t + (1 - \bar{\alpha}_t) = 1$ at every t . The process trades signal for noise at a fixed total — hence “variance preserving”.

The destination is free. $\alpha_s < 1$ for each s , so $\bar{\alpha}_t \rightarrow 0$ and $x_t \rightarrow \mathcal{N}(0, I)$. The forward process needs no training and its endpoint is a distribution we can sample perfectly. *That* is why diffusion models start by destroying the data.

7. Conditional expectation is what squared error computes

7.0 Notation: which distribution is each $\mathbb{E}$ taken over?

Every expectation below carries a subscript naming the distribution it integrates against. Without them the tower property looks circular — it appears to say $\mathbb{E}[\mathbb{E}[\cdot]] = \mathbb{E}[\cdot]$, which is either trivial or meaningless. It is neither, because *the three $\mathbb{E}$ ’s are three different operators*.

symbol	integrates over	the result is
$\mathbb{E}_{Y X=x}[\cdot]$	Y , with x held <i>fixed</i>	a number , depending on x
$\mathbb{E}_{Y X}[\cdot X]$	Y , with X left <i>symbolic</i>	a random variable : a function of X
$\mathbb{E}_X[\cdot]$	X	a number
$\mathbb{E}_{(X,Y)}[\cdot]$	both, jointly	a number

The concept

The one distinction that clears up the confusion.

$$\underbrace{\mathbb{E}_{Y|X=x}[Y]}_{\text{a number}} \quad \text{versus} \quad \underbrace{\mathbb{E}_{Y|X}[Y | X]}_{\text{a random variable}}$$

The second is what you get by *not* plugging in a value: it is the function $x \mapsto \mathbb{E}_{Y|X=x}[Y]$, regarded as a random quantity because its input X is random. Concretely: if X takes the values x_1, x_2, x_3 with probabilities p_1, p_2, p_3 , then the first object is one of the three numbers $m(x_1), m(x_2), m(x_3)$, chosen by which x_i you substituted — while the second object is the random variable taking those same three numbers with probabilities p_1, p_2, p_3 . It has a distribution of its own, which is precisely why averaging it is a further step and not a no-op.

The tower property then says something with content: averaging that random variable over X returns the plain average of Y ,

$$\mathbb{E}_X \left[\mathbb{E}_{Y|X}[Y | X] \right] = \mathbb{E}_Y[Y],$$

and more generally, for any h ,

$$\mathbb{E}_{(X,Y)}[h(X,Y)] = \mathbb{E}_X \left[\mathbb{E}_{Y|X}[h(X,Y) | X] \right].$$

Outer operator over X , inner operator over Y given X . They are not the same $\mathbb{E}$, which is why the statement is not a tautology.

Two facts used repeatedly below, both immediate once the subscripts are visible:

- **Pull-out.** If $a(X)$ depends on X alone, it is a *constant* for the inner operator: $\mathbb{E}_{Y|X}[a(X)W | X] = a(X)\mathbb{E}_{Y|X}[W | X]$.
- **Collapse.** If the integrand depends on X alone, the joint expectation is just the X -expectation: $\mathbb{E}_{(X,Y)}[a(X)] = \mathbb{E}_X[a(X)]$.

7.1 The best-predictor theorem

Claim. Among *all* functions g , the one minimising $\mathbb{E}_{(X,Y)}[(Y - g(X))^2]$ is $g(X) = \mathbb{E}_{Y|X}[Y | X]$.

Derivation. Define the function

$$m(x) = \mathbb{E}_{Y|X=x}[Y], \quad \text{so that} \quad m(X) = \mathbb{E}_{Y|X}[Y | X]$$

is a random variable depending on X only. Split the error around it:

$$Y - g(X) = (Y - m(X)) + (m(X) - g(X)).$$

Square and take the *joint* expectation — both terms involve X and Y , so this is $\mathbb{E}_{(X,Y)}$ throughout:

$$\mathbb{E}_{(X,Y)}[(Y - g)^2] = \mathbb{E}_{(X,Y)}[(Y - m)^2] + 2\mathbb{E}_{(X,Y)}[(Y - m)(m - g)] + \mathbb{E}_{(X,Y)}[(m - g)^2].$$

The cross term vanishes, and this is the only step that does any work. Apply the tower property to *split the joint expectation into two*:

$$\mathbb{E}_{(X,Y)}[(Y - m)(m - g)] = \mathbb{E}_X \left[\mathbb{E}_{Y|X}[(Y - m(X))(m(X) - g(X)) | X] \right].$$

Work on the *inner* operator only. It integrates over Y with X held fixed, so $m(X) - g(X)$ is a constant for it and pulls out:

$$\mathbb{E}_{Y|X}[(Y - m(X))(m(X) - g(X)) | X] = \underbrace{(m(X) - g(X))}_{\text{constant for } \mathbb{E}_{Y|X}} \underbrace{\mathbb{E}_{Y|X}[Y - m(X) | X]}_{= m(X) - m(X) = 0} = 0.$$

The inner expectation is zero *for every value of X* , so the outer $\mathbb{E}_X$ of it is zero as well. The third term involves X alone, so it collapses to $\mathbb{E}_X$, and the decomposition reads

$$\mathbb{E}_{(X,Y)}[(Y - g)^2] = \underbrace{\mathbb{E}_{(X,Y)}[(Y - m)^2]}_{\text{does not involve } g} + \underbrace{\mathbb{E}_X[(m(X) - g(X))^2]}_{\geq 0, \text{ zero only at } g=m}.$$

The first term is a floor you cannot get below — the noise in Y that X simply does not determine. The second is entirely under your control and is minimised, uniquely, at $g = m$. ■

The concept

Least squares does not fit a curve. It computes a conditional expectation.

Anything trained with a squared-error loss is estimating $\mathbb{E}_{\text{target}|\text{input}}[\text{target}]$ — whether or not the person who wrote the loss intended it. This is why a diffusion model trained on $\|\varepsilon - \varepsilon_\theta(x_t, t)\|^2$ ends up computing a posterior mean without any Bayesian machinery appearing in the code.

Free corollary — the law of total variance. Put $g \equiv \mathbb{E}_Y[Y]$, a constant function, into the same identity:

$$\underbrace{\mathbb{E}_Y[(Y - \mathbb{E}_Y Y)^2]}_{\text{Var}_Y(Y)} = \underbrace{\mathbb{E}_{(X,Y)}[(Y - m(X))^2]}_{\mathbb{E}_X[\text{Var}_{Y|X}(Y|X)]} + \underbrace{\mathbb{E}_X[(m(X) - \mathbb{E}_Y Y)^2]}_{\text{Var}_X(\mathbb{E}_{Y|X}[Y|X])}.$$

Read the two right-hand terms with their subscripts and the statement becomes concrete: the first averages, *over X* , the spread of Y that remains once X is known; the second measures how much the conditional mean itself moves *as X varies*. Variance splits into what observing X cannot explain plus what it can. One identity, two famous theorems.

7.2 What that theorem is worth

Take any latent Z you cannot observe and any measurement X you can. Compute the posterior $\mathbb{P}(Z \mid X = x)$ by Bayes and then its mean, by hand, for each x .

§7.1 says: **a network trained to predict Z from X under squared error converges to exactly those numbers.** Not approximately, and not as an artefact of the architecture — they are the minimiser. Bayes by hand and descent by machine arrive at the same function, which is why a diffusion model computes a posterior mean without a line of Bayesian code in it.

7.3 The same pipeline, with a Gaussian: the tanh denoiser

Now replace the table with a Gaussian and run Problem 5 again.

Setup. The clean signal is $x_0 \in \{-1, +1\}$, each with probability $\frac{1}{2}$ — the two-point dataset of note 01. We observe

$$x_\sigma = x_0 + \sigma \varepsilon, \quad \varepsilon \sim \mathcal{N}(0, 1),$$

so the likelihood is $p(x_\sigma \mid x_0) = \mathcal{N}(x_\sigma; x_0, \sigma^2)$. Same pipeline: prior, likelihood, posterior, conditional mean. As in §7.0, the averaging below is over the *posterior of x_0* at a fixed observed x_σ , which is what the subscript on $\mathbb{E}$ records.

Step 1 — the posterior. Bayes, with the prior $\frac{1}{2}$ cancelling top and bottom:

$$\mathbb{P}(x_0 = +1 \mid x_\sigma) = \frac{\exp\left(-\frac{(x_\sigma - 1)^2}{2\sigma^2}\right)}{\exp\left(-\frac{(x_\sigma - 1)^2}{2\sigma^2}\right) + \exp\left(-\frac{(x_\sigma + 1)^2}{2\sigma^2}\right)}.$$

Expand the squares, $(x_\sigma \mp 1)^2 = x_\sigma^2 \mp 2x_\sigma + 1$, and notice that $x_\sigma^2 + 1$ appears in **both** exponents. Divide top and bottom by $\exp\left(-\frac{x_\sigma^2 + 1}{2\sigma^2}\right)$ and it disappears, leaving only the cross terms:

$$\mathbb{P}(x_0 = +1 \mid x_\sigma) = \frac{e^{x_\sigma/\sigma^2}}{e^{x_\sigma/\sigma^2} + e^{-x_\sigma/\sigma^2}} = \text{logistic}\left(\frac{2x_\sigma}{\sigma^2}\right).$$

Step 2 — the conditional mean. With $p = \mathbb{P}(x_0 = +1 \mid x_\sigma)$ and values ± 1 , average over the posterior:

$$\mathbb{E}_{x_0|x_\sigma}[x_0] = (+1)p + (-1)(1-p) = 2p - 1 = \frac{e^u - e^{-u}}{e^u + e^{-u}}, \quad u = \frac{x_\sigma}{\sigma^2},$$

which is exactly the definition of the hyperbolic tangent:

$$\boxed{\mathbb{E}_{x_0|x_\sigma}[x_0] = \tanh\left(\frac{x_\sigma}{\sigma^2}\right).}$$

(The literature writes this $\mathbb{E}[x_0 \mid x_\sigma]$ and leaves the distribution implicit. That is safe *here*, because there is only one thing to average over; it is not safe in §7.1, where two nested operators are in play.)

That is the optimal denoiser for this dataset — the one from note 01 and from the W01 demo, now derived rather than asserted.

Read it against HW 1 Problem 5, line for line:

HW 1, Problem 5	Two-point diffusion
latent $Z \in \{1, 2\}$	latent $x_0 \in \{-1, +1\}$
observed $X \in \{0, 1, 2\}$	observed $x_\sigma \in \mathbb{R}$
prior (0.6, 0.4)	prior $(\frac{1}{2}, \frac{1}{2})$
likelihood: a 2×3 table	likelihood: $\mathcal{N}(x_\sigma; x_0, \sigma^2)$
posterior: three numbers	posterior: $\text{logistic}(2x_\sigma/\sigma^2)$
$\mathbb{E}_{Z X=x}[Z]$: three numbers	$\mathbb{E}_{x_0 x_\sigma}[x_0] = \tanh(x_\sigma/\sigma^2)$

Nothing changed except that the table became a formula.

Step 3 — the two limits, worth a minute each because they say what denoising *is*:

- $\sigma \rightarrow 0$: the argument x_σ/σ^2 blows up, $\tanh$ becomes a step function, and the denoiser confidently returns the nearer of ± 1 . Little noise, so trust the observation.
- $\sigma \rightarrow \infty$: the argument goes to 0 and so does $\tanh$. The denoiser returns 0 — the **prior mean** — because the observation carries nothing. It does not guess; it reports the average.

A denoiser interpolates between trusting the data and trusting the prior, and σ is the dial. Every diffusion model is doing this at every noise level simultaneously.

The concept

Optional — Tweedie’s formula. Stated, not proved:

$$\mathbb{E}_{x_0|x_\sigma}[x_0] = x_\sigma + \sigma^2 \nabla_x \log p_\sigma(x_\sigma).$$

The denoiser and the score are the same object in different clothes, which is the whole content of Module 4. You can check it here directly: differentiate $\log p_\sigma$ for the two-point mixture and confirm the right-hand side collapses to $\tanh(x_\sigma/\sigma^2)$.

8. KL divergence, and why it becomes a squared error

8.1 Definition, and non-negativity

$$\text{KL}(p \parallel q) = \mathbb{E}_p \left[\log \frac{p(X)}{q(X)} \right] = \int p(x) \log \frac{p(x)}{q(x)} dx.$$

Claim. $\text{KL}(p \parallel q) \geq 0$, with equality only when $p = q$.

Derivation by Jensen, using that $\log$ is concave so $\mathbb{E}[\log W] \leq \log \mathbb{E}[W]$:

$$-\text{KL}(p \parallel q) = \mathbb{E}_p \left[\log \frac{q(X)}{p(X)} \right] \leq \log \mathbb{E}_p \left[\frac{q(X)}{p(X)} \right] = \log \int p(x) \frac{q(x)}{p(x)} dx = \log 1 = 0.$$

Equality in Jensen for a strictly concave function requires the random variable to be constant, i.e. q/p constant almost surely — and since both integrate to 1, that constant is 1. ■

8.2 Between two Gaussians, derived

Let $p = \mathcal{N}(\mu_1, \sigma_1^2)$ and $q = \mathcal{N}(\mu_2, \sigma_2^2)$. Take the log-ratio first, since the exponentials make it easy:

$$\log \frac{p(x)}{q(x)} = \log \frac{\sigma_2}{\sigma_1} - \frac{(x - \mu_1)^2}{2\sigma_1^2} + \frac{(x - \mu_2)^2}{2\sigma_2^2}.$$

Now take $\mathbb{E}_p$ term by term. Two expectations are needed, **both under p** , which is where it goes wrong:

$$\mathbb{E}_p[(X - \mu_1)^2] = \sigma_1^2 \quad (\text{the variance of } p, \text{ by definition}),$$

$$\mathbb{E}_p[(X - \mu_2)^2] = \mathbb{E}_p[((X - \mu_1) + (\mu_1 - \mu_2))^2] = \sigma_1^2 + (\mu_1 - \mu_2)^2,$$

the cross term vanishing because $\mathbb{E}_p[X - \mu_1] = 0$. Substituting:

$$\text{KL}(\mathcal{N}(\mu_1, \sigma_1^2) \parallel \mathcal{N}(\mu_2, \sigma_2^2)) = \log \frac{\sigma_2}{\sigma_1} + \frac{\sigma_1^2 + (\mu_1 - \mu_2)^2}{2\sigma_2^2} - \frac{1}{2}.$$

Sanity check: set $\mu_1 = \mu_2$ and $\sigma_1 = \sigma_2$ and every term cancels to 0, as it must.

8.3 The collapse — the line Module 3 depends on

Set the two variances equal, $\sigma_1 = \sigma_2 = \sigma$. The log term is $\log 1 = 0$, and the middle term splits:

$$\frac{\sigma^2 + (\mu_1 - \mu_2)^2}{2\sigma^2} - \frac{1}{2} = \frac{1}{2} + \frac{(\mu_1 - \mu_2)^2}{2\sigma^2} - \frac{1}{2},$$

leaving

$$\boxed{\text{KL}(\mathcal{N}(\mu_1, \sigma^2) \parallel \mathcal{N}(\mu_2, \sigma^2)) = \frac{(\mu_1 - \mu_2)^2}{2\sigma^2}}$$

and in d dimensions with $\Sigma = \sigma^2 I$, the same computation gives $\|\mu_1 - \mu_2\|^2 / 2\sigma^2$.

The concept

A KL divergence between two Gaussians of equal width is a squared error, divided by a constant.

The DDPM training objective is a variational bound that decomposes into a sum of KL terms, one per noise level, each between two Gaussians that share a variance. By the boxed line above, every one of them is a squared error — which is why a derivation full of KL divergences ends at

$$\|\varepsilon - \varepsilon_\theta(x_t, t)\|^2.$$

There is no additional trick in Module 3. There is this identity, applied once per term.

8.4 Maximum likelihood is minimum KL

This closes the loop with HW 2, where you did MLE by hand without being told what it was minimising. Take the average log-likelihood and let n grow; by the law of large numbers,

$$\frac{1}{n} \sum_{i=1}^n \log p_\theta(x_i) \longrightarrow \mathbb{E}_{p_{\text{data}}}[\log p_\theta(X)].$$

Add and subtract $\mathbb{E}_{p_{\text{data}}}[\log p_{\text{data}}]$:

$$\mathbb{E}_{p_{\text{data}}}[\log p_\theta] = \underbrace{-H(p_{\text{data}})}_{\text{no } \theta} - \text{KL}(p_{\text{data}} \parallel p_\theta).$$

The entropy does not involve θ , so

$$\boxed{\arg \max_{\theta} \text{likelihood} = \arg \min_{\theta} \text{KL}(p_{\text{data}} \parallel p_\theta).}$$

Fitting was never “describe the data”. It was “get close to the data distribution” — we simply never had to name the notion of closeness, because likelihood chose one for us. Naming it is what exposes the choice, and the choice is what the rest of the course is about.

8.5 It is not symmetric, and here are the numbers

“KL is not a distance” is easy to accept verbally and then forget when reasoning. Kill that with arithmetic. Take $p = \mathcal{N}(0, 1)$ and $q = \mathcal{N}(0, \sigma^2)$ in §8.2:

$$\text{KL}(p \parallel q) = \log \sigma + \frac{1}{2\sigma^2} - \frac{1}{2}, \quad \text{KL}(q \parallel p) = -\log \sigma + \frac{\sigma^2}{2} - \frac{1}{2}.$$

At $\sigma = 2$:

direction	value
$\text{KL}(\mathcal{N}(0, 1) \parallel \mathcal{N}(0, 4))$	$\log 2 + \frac{1}{8} - \frac{1}{2} \approx 0.318$
$\text{KL}(\mathcal{N}(0, 4) \parallel \mathcal{N}(0, 1))$	$-\log 2 + 2 - \frac{1}{2} \approx 0.807$

Two and a half times apart, from the same pair of distributions. The asymmetry is not a technicality: $\text{KL}(p \parallel q)$ punishes q for being *narrow* where p has mass (mode-covering), and $\text{KL}(q \parallel p)$ punishes q for being *wide* where p has none (mode-seeking). Which order you write is a modelling decision, and note 09 §6 showed a GAN's practical loss quietly picking the second one.

Closing: what the session installed

Computation	Where it came from	Where it returns
Tower property, with subscripts	§7.0	every expectation argument in Module 3
$\mathbb{E}_{Z X}[Z \mid X]$ as a weighted average	§7.2	the denoiser output
Variances add	§6.2	the $\sqrt{t}$ envelope of the forward process
$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\varepsilon$	§6.3	training at a random t in one step
Squared error $\Rightarrow$ conditional expectation	§7.1	why the ε -loss is Bayesian
$\mathbb{E}_{x_0 x_\sigma}[x_0] = \tanh(x_\sigma/\sigma^2)$	§7.3	note 01, now derived
KL between equal-variance Gaussians	§8.3	the DDPM objective's final form
MLE = min KL	§8.4	why "which divergence?" organises the field

The concept

The one sentence. A diffusion model is a posterior over a hidden clean signal given a noisy observation, run at every noise level at once, with the table replaced by a Gaussian and the posterior mean computed by a network trained under squared error.

Everything in Module 3 is bookkeeping on top of that sentence.

Follow-on exercises

- A latent $Z \in \{1, 2\}$ has prior $(0.5, 0.5)$ and produces $X \in \{0, 1, 2\}$ with profiles $(0.8, 0.1, 0.1)$ and $(0.2, 0.2, 0.6)$. Compute the joint, the marginal $\mathbb{P}(X)$, the posterior $\mathbb{P}(Z=1 \mid X=x)$ and $\mathbb{E}_{Z|X=x}[Z]$ for each x , then verify the tower property. (*Every number is a clean fraction; if yours are not, check the joint sums to 1.*) Which of the two objects in §7.0 is each of your answers?
- Show $\mathbb{E}_X[\text{Var}_{Y|X}(Y \mid X)] \leq \text{Var}_Y(Y)$, and say when equality holds. (*What does equality mean about X ?*)
- Three noising steps: show directly that $x_3 = \sqrt{\alpha_1\alpha_2\alpha_3}x_0 + \sqrt{1 - \alpha_1\alpha_2\alpha_3}\bar{\varepsilon}$, without appealing to induction.
- With $\alpha_s = 0.99$ for every s , find the smallest t with $\bar{\alpha}_t < 0.01$. What does x_t look like at that point?
- Derive $\mathbb{E}[x_0 \mid x_\sigma]$ for a **three**-point dataset $x_0 \in \{-1, 0, +1\}$ with equal priors. Is it still a tanh?

- (f) For the two-point denoiser, compute $\text{Var}(x_0 \mid x_\sigma)$ and sketch it against x_σ . Where is the denoiser least certain, and does that match intuition?
- (g) Verify Tweedie's formula for the two-point mixture: show $x + \sigma^2 \nabla_x \log p_\sigma(x) = \tanh(x/\sigma^2)$.
- (h) Derive the Gaussian KL formula in d dimensions for $\Sigma_1 = \Sigma_2 = \sigma^2 I$ and confirm it is $\|\mu_1 - \mu_2\|^2 / 2\sigma^2$.
- (i) Compute $\text{KL}(\mathcal{N}(0, 1) \parallel \mathcal{N}(\mu, 1))$ and $\text{KL}(\mathcal{N}(\mu, 1) \parallel \mathcal{N}(0, 1))$. Why are these two equal when the variance case was not?
- (j) A network is trained with squared error to predict ε from x_t . Using §7.1, state exactly what function it converges to, and then use $x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\varepsilon$ to rewrite that function in terms of $\mathbb{E}_{x_0|x_t}[x_0]$. Say explicitly which distribution each expectation in your answer is taken over.